

Technical Document #959: Computer Vision - Comprehensive Overview

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Video understanding analyzes temporal dynamics in video sequences. 3D CNNs and transformer-based models capture spatiotemporal features.

Image classification assigns a label to an image from a fixed set of categories. CNNs like ResNet and EfficientNet achieve superhuman performance on benchmark datasets.

Semantic segmentation classifies each pixel in an image into a category. It is used in autonomous driving and medical image analysis.

Object detection identifies and localizes objects in images. YOLO and Faster R-CNN are popular real-time object detection architectures.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Key Takeaways:

- Visual question answering (VQA) answers questions about images.
- Image classification assigns a label to an image from a fixed set of categories.
- Face recognition identifies or verifies faces in images.